

UZAIR VAWDA

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Master CV — comprehensive historical record of roles, ventures, and projects. Reference document, not for submission. Source of truth for building targeted resumes. Last updated 2026-07-24.

PROFILE

Software engineer and MBA candidate operating at the intersection of technology strategy and execution, with a growing focus on AI-assisted and agentic engineering. Five-plus years across aerospace, financial services, and legal — building enterprise platforms, design systems, AI agents, and data-driven tools — alongside a portfolio of self-directed product ventures spanning AI/ML, mobile, and two-sided marketplaces. Produces both the strategy artifacts (PRDs, roadmaps, business cases, positioning) and the working software.

EDUCATION

Baruch College, Zicklin School of Business — Master of Business Administration

Expected June 2028

- Concentration: **Artificial Intelligence & Product Development.**

Drexel University, College of Computing & Informatics — B.S., Software Engineering

2021

- **GPA: 3.5 / 4.0.** Co-op program; early professional roles (Dechert, J.P. Morgan) completed as co-op placements.

PROFESSIONAL EXPERIENCE

Collins Aerospace

Nov 2022 – Present

Joined through the rotational Leadership Development Program; progressed across project management, IT strategy, and software engineering.

Software Engineer

Nov 2022 – Present

- Built and launched an internal **React/TypeScript design-system component library — 50+ reusable, accessible, themeable components** built on the Figma Simple Design System (SDS), open to all developers org-wide and **adopted by 5+ projects** to standardize the front-end and accelerate delivery.
- Built an in-house **AI agent (Poolside)** that recreates the design-system patterns and generates components on demand — cutting component-creation time by up to **80%** with copy-paste, props-ready snippets that ship with basic functionality built in; the agent has produced **5 custom, project-specific components** and is rolling out to new teams, retiring paid tooling for **~\$25K in savings to date.**
- Integrating the **AI agent (Poolside)** across broader development workflows to streamline engineering tasks and raise team productivity.
- Led a **platform-wide TypeScript migration** and modern-CSS (TailwindCSS) adoption, improving maintainability, consistency, and developer experience.
- Built **React/GraphQL** applications optimizing front-to-back data flow and integration across product teams.
- Owned end-to-end platform improvements — performance, testing infrastructure, observability, and CI/CD — reducing time-to-market.
- Partnered cross-functionally with design and product stakeholders to define requirements and set quality standards.

Infrastructure Project Manager, Leadership Development Program

Mar 2022 – Oct 2022

- Managed a portfolio of **6 infrastructure projects** across lifecycle stages, delivering 2 to completion while balancing competing priorities and stakeholders.
- Introduced Scrum practices that improved team velocity and cross-functional collaboration.
- Standardized company-wide PM artifacts (Project Charter, RACI matrix) adopted beyond the immediate team.

Applications Licensing Specialist, Leadership Development Program

Jul 2021 – Feb 2022

- Drove a software-license rationalization that cut license count **18%** and saved the business unit **\$66K+ year-over-year.**
- Consolidated **75+ applications across 6 servers**, improving accessibility and reducing redundancy.
- Built a React/Python license-tracking tool giving stakeholders real-time visibility into server status.

J.P. Morgan Chase & Co. — Front-End Experience Developer

Apr 2020 – Oct 2020

- Built reusable JavaScript/HTML5 components deployed across multiple sites, establishing shared front-end patterns.

- Delivered responsive, maintainable code against stakeholder requirements; used BitBucket and SonarQube for version control and code quality.

Dechert LLP — IT Applications Developer

Apr 2019 – Oct 2019

- Automated reporting and ETL workflows using SQL and Python for internal clients.
- Built Tableau dashboards surfacing application usage, performance, and adoption patterns.

ENTERPRISE PROJECT HIGHLIGHTS (WITHIN COLLINS AEROSPACE)

RFCC — Enterprise React Component Library / Design System

- Shared, themeable React component library ("Rotor Figma Code Connect") built on Figma's Simple Design System (SDS); **50+ accessible components** backed by CSS custom-property design tokens.
- Packaged and published to the internal Nexus npm registry with a live component dashboard, Storybook, a standalone consumer demo, and a documented publish pipeline (Jenkins + SVN + Nexus).
- Open to all developers org-wide; **adopted by 5+ projects** with adoption actively growing.
- Themes: light/dark + brand overrides via two-layer token system; design-to-code workflow via Figma Code Connect.
- Built an in-house **AI agent (Poolside)** to recreate SDS patterns and auto-generate components — cutting component-creation time by up to **80%** via copy-paste, props-ready snippets with basic functionality built in; **5 custom, project-specific components** shipped and rolling out to new teams, retiring paid tooling for **~\$25K in savings to date**; expanding into broader development workflows.
- **Themes/skills:** enterprise design systems, internal-platform productization, AI-assisted development, developer experience, design-to-code, accessibility, cross-team stakeholder coordination.

Skyler — Aerospace/Defense Sensor Platform UI

- Angular site-configuration interface for a radar/sensor platform: live telemetry visualizations (array tilt, azimuth compass, pitch/roll attitude indicators, MapLibre GL mapping) and full radar **scan-mission management** (STARE, PPI, RHI, raster patterns).
- Rigorous phased product spec (data models, API contracts, behavior tables, responsive rules); built as a standalone module with documented integration seams for a larger production system.
- **Themes/skills:** complex-domain product design, requirements discipline, systems-integration mindset, data visualization (Angular, RxJS, Angular Material, MapLibre GL, SVG).

VENTURES & ENTREPRENEURSHIP

JHParking — Founder / Full-Stack Developer

June 2025 – Present

- **Peer-to-peer urban parking marketplace** ("Airbnb for parking") letting owners list and rent personal spaces short-term.
- React/TailwindCSS front end, **Stripe** payments, **Firestore** backend; two-sided marketplace.
- In active development, **currently piloting with 7 beta users** (excl. founder); scoped intentionally as a focused local solution rather than a mass-market app.
- Evolved from an earlier prototype (**ParkForLess**) — multi-year conviction on the idea.

MatAI — Founder / Product & Technical Design (*in development*)

July 2026 – Present

- Designing an **AI video-analysis system for Brazilian Jiu-Jitsu** that ingests up to 6-minute match footage and outputs a **timestamped positional timeline, submission-attempt detection, and coaching feedback. Pilot planned with roughly 40 volunteer testers** from a jiu-jitsu training team once the first build is ready.
- **Five-stage pipeline architecture:** automated footage-quality probing (resolution, Laplacian-variance sharpness scoring, camera-stability detection, athlete-visibility check) that gates and caps downstream confidence; ffmpeg-based adaptive frame extraction; concurrent analysis of overlapping 45-second windows against a vision-language model; deterministic timeline stitching with overlap reconciliation and contradiction detection; and a text-only synthesis stage for cross-roll pattern recognition. **Stack:** Python, FastAPI, ffmpeg, Pydantic, SQLite, Anthropic Claude API (Claude Opus 5), vanilla JS front end with zero build step.
- **Chose a relational-state architecture over conventional multi-object tracking.** Standard approaches (YOLO + ByteTrack, pose estimation, SAM 2 segmentation) break under the near-total mutual occlusion of grappling, where bounding boxes overlap almost completely and trackers permanently swap athlete identities. Since jiu-jitsu positions are inherently relational — "closed guard" describes a two-body relationship, not an individual — the design classifies the **joint dyad state plus role assignment** (top/bottom, guard-player/passers) instead of tracking two independent entities, binding roles to athletes via appearance descriptors extracted once at probe time.

- **Designed for graceful degradation under uncertainty:** a **30-position hierarchical taxonomy** with family-level fallback; "scramble" and "unclear" as **first-class output labels** to eliminate forced-guess confabulation; adaptive re-sampling that re-extracts high-motion windows at 6 fps when scramble density or mean confidence crosses threshold; structured JSON enforced through schema constraints; prompt-prefix ordering engineered to maximize prompt-cache hit rate across windows.
- **Specified the evaluation framework before writing the system:** hand-labeled ground-truth corpus of 10-20 rolls; per-second timeline accuracy at both specific-position and position-family granularity; submission-attempt precision and recall within a 2-second matching tolerance; hallucination rate on high-confidence claims; confidence-calibration curves; and a standing policy that **no prompt change ships without a measured before/after delta**. Ranked precision and hallucination-rate targets **above raw recall** on the reasoning that a single fabricated finding destroys user trust in the correct ones.
- **Cost and latency engineered as first-class constraints:** per-stage token usage, cost, and wall-clock latency instrumented from the first build, targeting a **\$5-per-video budget ceiling** and a **5-minute processing SLA**, with a documented degradation ladder (sampling rate, model tier, frame resolution, window width) to be exercised only after the accuracy ceiling is measured.
- **Themes/skills:** applied AI product engineering, problem reframing, pipeline architecture, evaluation design and calibration, cost/latency engineering, ambiguity handling as product design, 0-to-1 venture definition.

CoachMe — Founder / Product & Technical Design (*in development; documentation stage*)

- Designing a **two-sided marketplace** where competitive jiu-jitsu athletes buy **video review from vetted high-level coaches**. **5 coaches committed**; pre-launch, with no athletes or orders yet.
- Feedback is **anchored to the footage itself**: notes pinned to exact timestamps, recorded voiceover against the timeline, vector **telestration on paused frames in normalized coordinates** so drawings scale across devices, and scores against a **versioned rubric**.
- **Replaced a fixed product catalog with entitlement primitives** — coaches compose their own offerings (review quantity, permitted feedback modes, messaging level, live calls, training plan, revision count), set their own price, and set their own turnaround SLA under a **7-day platform ceiling**.
- **Resolved the conflict between coach differentiation and longitudinal athlete tracking by separating what is measured from what it is worth**: the platform owns a fixed versioned rubric dimension set that every review scores; **coaches own the importance weights**, published on their profile as their coaching philosophy and searchable as a discovery axis; coach-authored extension dimensions are scored in-review but **excluded from cross-coach progression charts**, so an athlete switching coaches never sees a fake cliff in their data.
- **Architected the money layer on Stripe Connect Express using separate charges and transfers rather than destination charges**, specifically so funds can be held in **platform escrow** between purchase and acceptance — destination charges settle immediately and leave nothing to hold, which would make the SLA and dispute process unenforceable.
- Specified a **12-state order machine** (requested, accepted, in_progress, delivered, completed, plus declined, expired, disputed, arbiter_review, and refunded paths) as a **pure, total transition function** in a framework-free domain package consumed identically by the Next.js web app, the Expo mobile app, and the background job service; **illegal transitions return typed errors** rather than throwing or silently succeeding, and the state machine performs **no I/O**.
- **Confined all fund movement to idempotent Inngest workflows** — never a request handler, never a cron scanning stale rows — with every state change writing an **append-only ledger entry**. Delivery deadlines compute as **min(coach SLA, acceptance + 7 days)**; a breach **auto-refunds in full with no human in the loop**, keeping the high-volume zero-judgment failure case out of the dispute queue and making an **unpaid volunteer arbiter pool** operationally viable.
- **Modeled unit economics against real Stripe and Mux pricing** to land a **flat 10% commission** — below the 15-20% creator-marketplace norm, chosen because **supply is the binding constraint** — and derived a **\$40 minimum offering price** from the point where Stripe's fixed \$0.30 per-charge fee makes an order unprofitable.
- **Put authorization in Postgres rather than the application layer**: every table ships with **row-level security** and a **default-deny** posture, with a **pgTAP policy test** proving both that the entitled role can read and that every other role cannot, so a mistake in an API route cannot leak another athlete's footage. No video asset is ever public — playback requires a **server-minted signed Mux JWT with a TTL of 15 minutes or less**, issued only after an access check and never logged or persisted.
- **Designed the dispute-arbiter boundary around the risk that an always-on third-party observer would chill honest conversation about injuries and weaknesses**: arbiters are reachable through support from day one but hold **zero standing data access**; reading an order requires a **grant scoped to that single order** — time-boxed, revocable, immutably audit-logged, and visible to both parties — backed by a **recusal duty**, since arbiters come from a real training community.

- Additional controls specified: **MFA on coach and admin roles as a hard gate on Stripe Connect onboarding**, so no account can be positioned to receive money without it; chat attachments **quarantined behind a malware scan** before any signed URL issues; PostHog session replay **masking video surfaces and payment inputs**.
- **Identified storage as a permanent liability against one-time commission revenue** — at sufficient order volume the storage bill on old footage exceeds the commission on new footage — and designed three controls that **do more for margin than pricing does**: a **1080p ingest cap** with downscale at encode (4K storage costs 3.2x and phones default to it), **mandatory athlete-side trimming** before submission, and **automatic deletion of untrimmed sources and never-ordered footage**. Modeled to cut a year of retention from more than the full commission on the order to **roughly 14% of net**, which is what makes a 12-month included retention window and a flat annual archive tier viable.
- **Themes/skills**: two-sided marketplace design, take-rate strategy under supply constraint, product judgment across conflicting goals, payments and escrow architecture, state-machine and domain modeling, security and privacy engineering (RLS, signed media, MFA gating), trust-and-safety design, unit economics and cost engineering.

PageKeeper — Product Management & Design (*self-directed concept*)

- End-to-end mobile product spec (React Native / Firebase) for a reading-habit app: streaks, goals, gentle friend accountability, reading journal.
- Complete PRD with personas, data models, a full **security & compliance architecture** (encryption, auth, Firestore rules, rate limiting, GDPR/CCPA/COPPA), ethical monetization strategy, and success metrics.
- **Themes/skills**: end-to-end PM, security/privacy/compliance fluency, metrics-driven design.

MBA-Engineered — Consulting Portfolio Strategy (*concept/positioning*)

- Personal-brand strategy: ship one small consulting tool per MBA class (IT maturity assessment, market-sizing dashboard, Porter's Five Forces analyzer, an "Ask My MBA" **RAG** app, etc.), positioning as someone who can both advise on and build consulting deliverables.

Connect. — Influencer Marketing Platform (*concept/prototype*)

- Vue-based concept for an all-in-one platform connecting companies and influencers to optimize reach — early product/business framing.

LEADERSHIP & LIVE EVENTS

Muslim Interscholastic Tournament (MIST) — Finance Coordinator

2017 – 2021

- **Owned the tournament budget** across four tournament cycles, managing spend against sponsorship revenue and **growing the budget 5–20% year over year**.
- **Secured and managed corporate sponsors** (including **PwC**) — owned outreach, sponsor deliverables, and the sponsor relationship.
- Planned and ran **live interscholastic competition events end-to-end** — program schedule and run-of-show, speaker and judge coordination, venue and vendor management, volunteer staffing, and day-of execution across concurrent competition tracks.
- **Themes/skills**: budget ownership, sponsorship revenue, vendor management, live event production, run-of-show, speaker coordination, high-pressure day-of execution.

SELECTED INDEPENDENT & ACADEMIC PROJECTS

Marketplaces & Products

- **ParkForLess** — "Airbnb for parking" prototype (React, Firebase, Leaflet). Precursor to JHParking.
- **718SNKRS** — Sneaker e-commerce store with auth and payments (Express, MongoDB, Stripe, sessions).
- **Reddit Clone** — Full Reddit-style app (Next.js, TypeScript, Chakra UI, Firebase, Recoil).

Web & Full-Stack

- **DragonFeed** — Drexel campus events/news aggregator (academic project).
- **Curriculum App** — Course/curriculum tool (Vue + Node) — INFO 420 project.
- **Project Tracker** — Project-tracking app (Vue, Firebase, Vuetify).
- **ExpressBlog** — Blogging platform with auth (Express, MongoDB, EJS).
- **ChatApp** — Real-time chat (MERN stack).
- **Dynamic Site / Hosting** — Server-rendered dynamic site (Express, EJS, MongoDB).
- **Web Portfolio** — Personal site, uzairvawda.me.

Automation & Data

- **Send a Script** — Python automation to deliver movie scripts via iMessage / Facebook Messenger / WhatsApp.
- **Investment Tracker (invs)** — Trade history, monthly rollups, and metrics/visualizations for personal investing data.

SKILLS

Strategy & Advisory: Digital transformation · Technology & product strategy · Business case development · Cost optimization & process improvement · Market & competitive analysis · Go-to-market (GTM) strategy · Operating-model & requirements analysis · Change management · Stakeholder management

Product Management: Product roadmapping · PRD authoring · MVP scoping & prioritization · User personas & user research · SaaS & two-sided marketplace monetization · Unit economics & pricing · Marketplace take-rate design · Trust & safety · KPIs & success metrics · A/B testing · Agile / Scrum (CSM, CSPO) · PMP (expected Dec 2026) · RACI · Cross-functional leadership

AI & Data: AI/ML product design · AI-assisted / agentic development · Computer vision · Vision-language models · Large language models (LLMs) · Prompt engineering · Model evaluation & benchmarking · Retrieval-augmented generation (RAG) · Generative AI · Data analysis & visualization · Tableau · SQL · Dashboards

Engineering & Platform: TypeScript, JavaScript, Python, C#, SQL · React, Vue, Next.js, Node, Express · GraphQL & REST APIs · Design systems & component libraries · Firebase/Firestore, PostgreSQL, MongoDB, MySQL · Stripe / Stripe Connect · CI/CD · Cloud · Git · Jira · Excel · Google Slides

CERTIFICATIONS

Project Management Professional (PMP) — Project Management Institute	Expected Dec 2026
Certified ScrumMaster (CSM) — Scrum Alliance	Apr 2022
Certified Scrum Product Owner (CSPO) — Scrum Alliance	Oct 2022
100 Days of Front-End Development — Udemy	Aug 2022

ADDITIONAL

Languages: English · Urdu · Gujarati · Conversational Spanish

Activities: Muslim Interscholastic Tournament (MIST) — Finance Coordinator, 2017–2021 · Muslim Student Association · Drexel Coding Club · Powerlifting